

QUANG NGAI, Viet Nam

WMO: 488630

Lat: 15.133N

Lon: 108.783E

Elev: 30

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	63.9	65.3	57.7	71.1	66.8	59.5	76.0	67.3	11.2	71.5	9.6	72.0	3.1	320	0.258

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	13.4	96.9	81.6	95.3	81.0	94.0	80.6	82.8	93.3	82.2	92.2	81.6	91.0	4.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.2	157.1	87.4	79.7	154.4	87.0	79.1	151.4	86.4	46.9	94.1	46.1	92.7	45.5	91.5	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.4	7.4	61.6	98.9	2.0	2.1	60.2	100.4	59.0	101.7	57.9	102.8	56.5	104.4		
(5)			58.8	83.6	1.6	1.4	57.6	84.6	56.6	85.4	55.7	86.2	54.5	87.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.1	72.3	74.0	77.2	81.7	85.2	86.5	85.6	85.1	82.8	79.8	77.3	73.6
	DBStd	5.77	3.36	3.76	3.94	3.02	2.72	2.42	2.39	2.71	2.57	2.26	3.15	3.61
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10998	692	672	845	951	1092	1095	1103	1089	984	923	820	732
	CDD65	5523	227	252	380	501	627	645	638	624	534	458	370	267
	CDH74	58054	741	1378	3037	5483	8127	8744	8349	7984	6123	4227	2729	1132
	CDH80	23304	53	213	772	2047	3900	4528	4062	3766	2340	1035	482	106
Wind	WSAvg	3.1	3.0	3.4	3.6	3.3	3.0	2.8	2.7	2.6	2.8	2.9	3.5	3.2
Precipitation	PrecAvg	107.2	4.4	1.4	2.0	1.4	3.1	3.0	3.1	5.9	11.9	27.0	23.5	14.0
	PrecMax	159.6	11.3	4.4	8.7	12.0	9.3	11.7	8.7	21.5	30.0	44.8	56.4	52.4
	PrecMin	72.0	0.6	0.2	0.2	0.0	0.4	0.6	0.9	1.4	2.5	10.9	5.2	1.4
	PrecStd	25.9	2.8	1.1	2.0	2.4	2.3	2.7	2.1	4.4	6.9	10.6	13.9	12.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.9	86.5	89.1	94.4	98.1	99.1	98.7	98.1	95.0	90.0	88.1	85.2
		MCWB	76.4	77.2	78.7	81.2	82.1	82.7	81.6	81.4	81.9	79.8	79.3	76.6
	2%	DB	81.3	84.5	87.3	91.8	95.6	96.6	95.7	95.7	92.6	88.1	85.6	82.7
		MCWB	74.2	75.9	77.8	80.2	81.4	81.6	80.6	81.1	80.9	78.8	77.8	76.1
	5%	DB	79.6	82.6	85.8	89.9	94.2	95.0	93.8	93.8	90.7	86.4	84.4	80.8
		MCWB	73.2	74.8	77.2	79.4	80.9	80.9	80.2	80.5	80.1	78.0	77.3	75.0
	10%	DB	77.9	80.9	84.3	88.3	92.4	93.4	92.2	91.8	89.0	85.0	82.9	79.1
		MCWB	72.3	73.8	76.4	78.6	80.4	80.5	79.9	80.0	79.5	77.5	76.7	74.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	78.0	80.1	82.2	83.5	83.9	83.3	83.6	82.8	81.1	80.1	78.1
		MCDB	82.0	84.9	86.8	92.0	94.8	95.9	93.9	94.3	92.1	87.5	85.6	82.0
	2%	WB	75.5	76.7	78.9	80.9	82.3	82.7	82.3	82.3	81.7	79.9	79.0	76.9
		MCDB	79.4	82.4	85.5	89.6	93.1	93.3	92.1	92.3	90.3	85.6	83.8	80.8
	5%	WB	74.5	75.6	77.9	80.1	81.6	81.9	81.5	81.5	80.8	79.1	78.2	76.0
		MCDB	78.0	80.7	84.0	87.9	91.5	91.8	90.7	90.7	88.3	84.3	82.5	79.6
	10%	WB	73.5	74.7	77.0	79.3	80.9	81.2	80.7	80.7	80.0	78.4	77.5	75.1
		MCDB	76.7	79.3	82.5	86.4	89.9	90.5	89.4	89.4	86.9	83.1	81.4	78.3
Mean Daily Temperature Range	5% DB	MDBR	7.6	9.8	10.6	11.9	13.3	13.4	12.9	12.7	11.0	8.8	7.5	6.8
		MCDBR	10.6	12.9	13.2	14.5	16.0	15.8	15.4	15.5	14.2	11.7	10.6	9.7
		MCWBR	4.4	5.4	5.3	5.0	5.0	4.8	5.1	5.2	5.2	4.3	4.0	4.2
	5% WB	MCDBR	9.0	12.1	12.3	13.6	14.9	14.8	14.2	14.5	13.1	10.1	9.2	8.3
		MCWBR	4.4	5.4	5.1	5.1	5.3	5.1	5.5	5.5	5.2	4.3	4.1	4.1
Clear-Sky Solar Irradiance	taub	0.472	0.487	0.581	0.545	0.478	0.453	0.438	0.442	0.476	0.522	0.507	0.517	
	taud	2.189	2.160	1.925	2.066	2.290	2.374	2.405	2.384	2.289	2.117	2.130	2.044	
	Ebn at Noon	260	262	241	249	262	266	271	272	264	249	247	241	
	Edh at Noon	45	48	62	54	43	39	38	39	43	50	48	51	
All-Sky Solar Radiation	RadAvg	1090	1446	1609	1846	1887	1816	1730	1677	1543	1268	1079	902	
	RadStd	135	208	140	123	118	110	145	143	161	142	187	175	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.59	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	+410	+407

Nomenclature: See separate page